

SwipeX

1. Title: SwipeX: Swipe-Based Intelligent Job Discovery and Career Assistance Platform

2. Objective

Build a swipe-based intelligent job discovery and career assistance platform that helps candidates discover and apply for relevant jobs using AI-powered resume analysis, ATS scoring, personalized recommendations, and real-time opportunity tracking.

The system should allow users to explore opportunities through an intuitive swipe-based interface, analyze resume compatibility, receive personalized recommendations, track applications, and identify the most suitable opportunities across MNCs, startups, and newly founded companies.

The platform is designed to support modern job discovery by combining artificial intelligence, recommendation systems, ATS optimization, and career analytics into a centralized application.

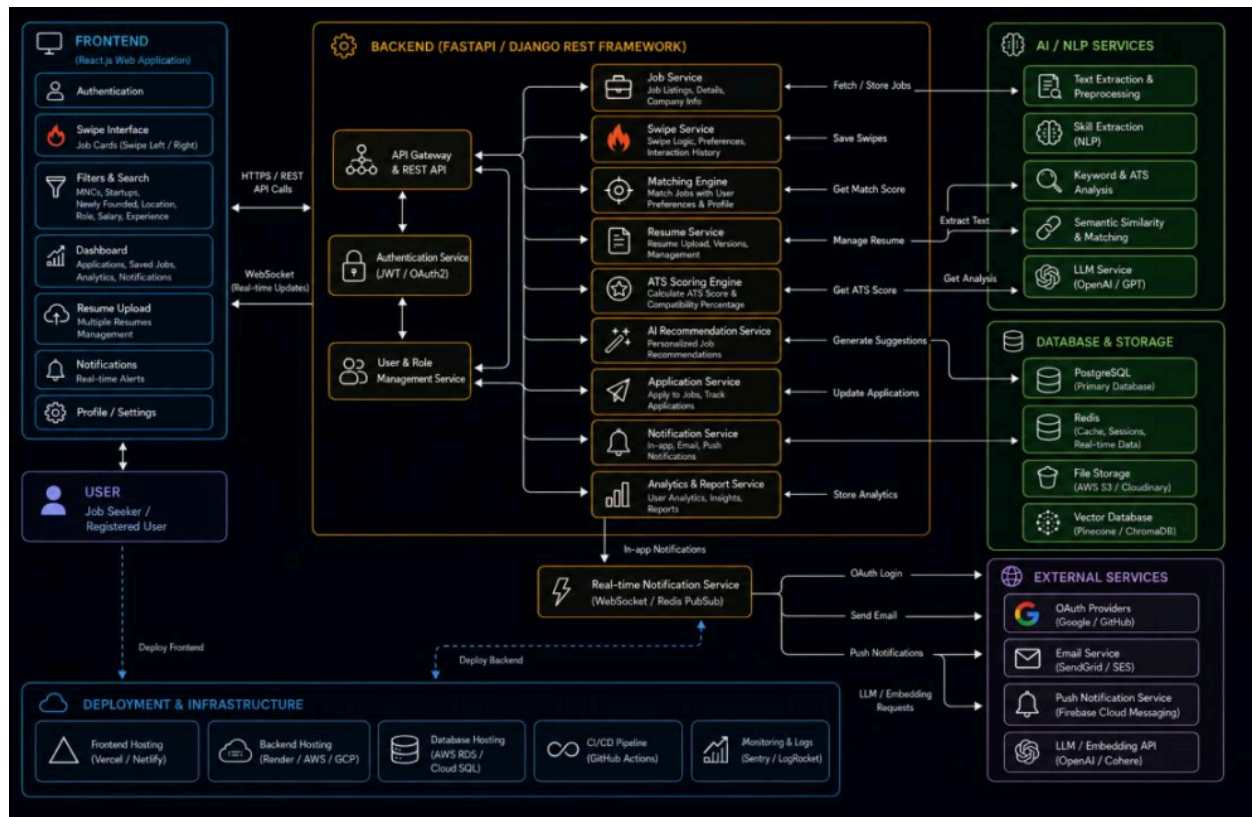
This solution can be used by students, fresh graduates, experienced professionals, recruiters, career coaches, and job placement platforms.

Outcomes

- Developed and deployed a swipe-based intelligent job discovery platform.
- Implemented secure authentication and role-based access control.
- Built swipe-based job exploration and application workflows.
- Integrated AI-powered resume analysis and ATS scoring.
- Developed personalized recommendation and matching systems.
- Implemented real-time job freshness and competition tracking.
- Built dashboards for applications, analytics, and recommendations.
- Added startup discovery and intelligent notification systems.

- Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.

3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

Authentication Features

- JWT Authentication
- OAuth2 Login
- Secure Session Management

Role Management

Job Seeker

- Create and manage profiles
- Upload resumes
- Apply for jobs
- Save jobs
- Track applications
- View recommendations

Recruiter

- Post jobs
- Review applicants
- Manage hiring workflows
- Access analytics
- Monitor application trends

Admin

- Manage users
- Manage recruiters
- Monitor platform activities
- Configure system settings
- Access platform-wide analytics

2. User Profiles & Resume Management

- Profile creation
- Resume upload
- Multiple resume versions
- Skill tagging
- Experience management

- Portfolio management

3. Swipe-Based Job Discovery

Swipe Actions

Swipe Right:

- Apply
- Save jobs
- Add to favorites

Swipe Left:

- Skip jobs
- Improve future recommendations

Features

- Dynamic job cards
- Personalized recommendations
- AI-powered preference learning

4. Smart Job Search & Filtering

Users can filter jobs by:

- MNCs
- Startups
- Newly founded startups
- Remote jobs
- Internship opportunities
- Full-time jobs
- Salary range
- Skills and technologies

- Location
- Experience level

Advanced Filters

- Fresher-friendly jobs
- Low competition jobs
- Recently posted jobs

5. Real-Time Job Freshness & Competition Tracking

- Posted-time indicators
- Applicant count display
- Competition level tracking
- Early applicant indicators

Competition Levels

- Low
- Medium
- High

6. AI Resume Analyzer & ATS Scoring

- Resume upload and parsing
- ATS compatibility analysis
- Keyword matching
- Missing skill detection
- Resume-job compatibility scoring
- AI-generated improvement suggestions

Example

If ATS Match Score is below 80%:

- Missing skills are highlighted
- Missing keywords are identified
- Resume optimization recommendations are generated

7. Personalized Job Recommendation System

- Skill-based recommendations
- Resume-based recommendations
- Experience-based suggestions
- AI-generated match percentages
- Recommendation learning from swipe behavior

8. Smart Notifications & Alerts

- Instant job notifications
- Startup hiring alerts
- Personalized recommendations
- High-match opportunity alerts
- Low-competition opportunity alerts

9. Application Tracking System

- Saved jobs tracking
- Applied jobs tracking
- Interview tracking
- Rejected status tracking
- Shortlisted status tracking
- Application history dashboard

10. Dashboard & Analytics

- Application analytics

- Resume performance tracking
- Skill gap analysis
- Recommendation insights
- Hiring trend visualization

11. Final Integration, Testing & Deployment

- Module integration
 - End-to-end testing
 - Performance validation
 - Production deployment
 - Documentation and demonstration
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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Core Setup

- Define project scope and user roles.
- Design database schema.
- Setup frontend and backend environments.
- Configure PostgreSQL database.
- Implement authentication APIs.
- Build initial frontend structure.

Outcomes

- Backend and frontend architecture setup completed.
- Authentication workflows functional.
- Database schema finalized.

Milestone 2: Week 3 & 4 — Swipe System & Job Discovery

- Build job listing APIs.
- Implement swipe-based recommendation workflows.
- Create filtering and search functionality.
- Develop job card interfaces.
- Build company and startup listing pages.

Outcomes

- Swipe-based job discovery operational.
- Dynamic recommendation feed implemented.
- Smart filtering system functional.

Milestone 3: Week 5 & 6 — AI Resume Analysis & Recommendations

- Implement resume upload and parsing workflows.
- Build ATS scoring engine.
- Develop resume-job compatibility analysis.
- Implement recommendation algorithms.
- Generate AI-powered suggestions.

Outcomes

- AI resume analyzer operational.
- Match percentage system functional.
- Resume optimization recommendations generated.

Milestone 4: Week 7 & 8 — Notifications, Analytics & Deployment

- Implement notification systems.
- Build analytics dashboard.
- Add applicant competition indicators.

- Perform testing and validations.
- Deploy platform using cloud environments.
- Configure environment variables and security settings.

Outcomes

- Fully deployed production-ready application.
 - Notification and analytics systems operational.
 - Complete end-to-end workflow demonstrated.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Backend initialization completed.
- Authentication workflows implemented.
- Database schema finalized.
- Frontend setup completed.

Milestone 2 (Week 4)

- Swipe-based job discovery operational.
- Smart filtering system functional.
- Dynamic recommendation feed implemented.

Milestone 3 (Week 6)

- Resume upload and ATS scoring operational.
- AI recommendation system functional.
- Resume-job compatibility workflow completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.

- Notification and analytics systems operational.
 - End-to-end workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend: Python (Django REST Framework / FastAPI)

Frontend: React.js

Database

- PostgreSQL
- SQLite (Development)

AI & NLP Technologies

- OpenAI API
- spaCy
- sentence-transformers
- scikit-learn

Authentication

- JWT Authentication
- OAuth2 Authentication

Cloud & DevOps

- Docker
- AWS / Azure
- GitHub Actions

Libraries & Frameworks

- Django

- Django REST Framework
- FastAPI
- React.js
- React Router
- Axios
- Tailwind CSS
- Redux / Context API
- Framer Motion

Testing Tools

- Pytest
- Django Test Client
- Jest
- React Testing Library

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - GitHub Actions
 - Postman
 - Render / AWS / Vercel
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8. Performance Metrics

Job Recommendation Performance

- Recommendation accuracy
- Job matching relevance

- ATS scoring effectiveness
- Recommendation response time

Resume Analysis Performance

- Resume parsing accuracy
- Skill extraction accuracy
- Keyword detection effectiveness
- Resume-job compatibility accuracy

Analytics Performance

- Dashboard response time
- Report generation performance
- Recommendation processing efficiency

System Performance

- API response time
 - Concurrent user handling
 - Database query optimization
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9. Example Quantitative Goals

Job Discovery

Provide highly relevant job recommendations based on candidate profiles and resume analysis.

Resume Optimization

Generate actionable ATS and resume improvement suggestions.

Recommendation Engine

Improve job matching accuracy through personalized recommendation algorithms.

Analytics

Provide real-time insights into applications, recommendations, and hiring trends.

Platform Performance

Support multiple concurrent users with stable recommendation and application tracking workflows.